

Derek Su

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| [Github](#) | [Linkedin](#)

EDUCATION

University of California, Berkeley

Bachelor of Science in Electrical Engineering and Computer Science

Berkeley, CA

May 2028

WORK EXPERIENCE

Northrop Grumman

Software Engineering Intern (Interim Security Clearance)

Redondo Beach, CA

June 2026 - August 2026

- Accelerated high-latitude terrain analysis by up to **36%** by doubling parallel processing from **4 to 8 threads**, improving slope-generation performance.
- Automated ATO/ACO conversion for mission-planning workflows, reducing manual conversion time by **30%** with a Python engine mapping 243 schema-defined fields from USMTF 2004 to 2000 per MIL-STD-6040.

Whitehelmet

Software Engineering Intern

Remote

Feb 2026 - June 2026

- Reduced manual data reconciliation by **63%** across **20** organizations by building an AI-powered pipeline that normalized inconsistent Excel templates into standardized KPI reports.
- Reduced report generation time by **80%** by developing a server-side Excel engine supporting **10+** configurable KPI formulas and LLM-powered schema reconciliation.
- Architected **91 REST APIs** with FastAPI, SQLAlchemy, and PostgreSQL for multi-tenant reporting, template management, spreadsheet automation, and role-based access control.

Lucid Motors

Software Developer (Contract)

Berkeley, CA

Sep 2025 - December 2025

- Reduced manual competitor research time by **87.5%** by building an AI-powered automotive intelligence platform that automated pricing collection and analysis across **8** competitor brands.
- Reduced analytics latency by **90%** and improved rendering speed by **30%** by building modular data-processing agents for the automotive intelligence pipeline.
- Shipped **10+ production features** across data collection, storage, dashboard analytics, and AI querying while collaborating with a **6-person** engineering team through weekly releases.

Claythis

AI/ML Intern

San Francisco, CA

Feb 2025 - June 2025

- Owned experimentation for an AI character-generation pipeline, evaluating **5+ diffusion models** across **20+** test cases and achieving **90%+** estimated visual similarity to reference characters.
- Improved pose consistency by **~70%** by building a multimodal conditioning pipeline combining ControlNet and OpenPose for pose-guided video generation.
- Reduced model iteration time from **10+ minutes to ~2-3 minutes** by benchmarking RunPod GPU configurations across generation quality, latency, and cost.

PROJECTS

[Google Docs AI Editor](#) - a16z x Berkeley M.E.T Hackathon, 1st out of 200 teams

Sep 2025

- Built a Google Docs AI editor integrating Claude with **2 editing modes** and up to **5 attachments**, enabling document-aware generation and reviewable Accept/Reject edits without leaving the editor.
- Engineered a range-based editing system supporting reliable selections, cursor insertions, deletions, and multi-paragraph changes while preserving document integrity and reversible AI edits.
- Achieved **~2-second average AI response latency** with a serverless Google Apps Script architecture using **5 asynchronous RPC endpoints** and document-scoped state management at **\$0 hosting cost**.

[Real-Time League of Legends AI Coach](#) - CalHacks

November 2025

- Architected a multimodal real-time coaching pipeline integrating **4 gameplay signals**—OCR, Riot APIs, audio, and voice—to deliver AI recommendations with **<3-second end-to-end latency**.
- Achieved **90% real-time ability-detection accuracy** using FFT-based audio matching with SciPy and PyAudio, complementing visual and API signals with audio events.
- Delivered **20+ real-time coaching events per minute** by designing a priority-based engine that reconciled OCR, Riot API, audio, and LLM signals into actionable recommendations.

SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, C++, SQL, MATLAB

Frameworks & Libraries: React, Next.js, FastAPI, Node.js, PyTorch, TensorFlow, Hugging Face Transformers

Cloud & DevOps: AWS (EC2/S3), GCP, Docker, Kubernetes, Vercel, Supabase (PostgreSQL), Redis, Git, CI/CD

AI Development: Codex, Claude Code, MCP, Agent Skills, Hooks & Automation